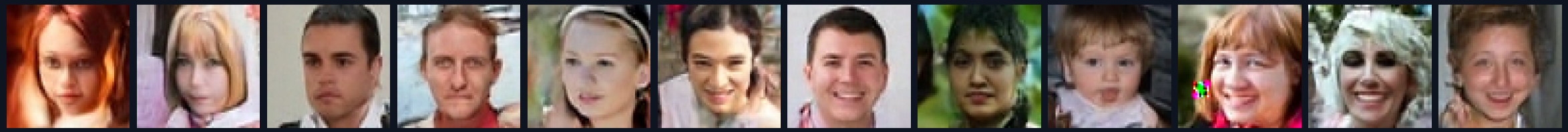
Frequency-AR image generation: what have we learned?



A1 (ours, 11.1M) · 250k EMA · UniPC [20,10,8,8] · std seeds 0–11, uncurated — seed 9 shows the rare single-token G3 checkerboard (slide 10)

THE MODEL

$$p(X) = \prod_k p(G_k \mid G_{<k})$$



G0	DC	UniPC 20 calls
G1	(0,1) (1,0)	UniPC 10 calls
G2	(2,0) (1,1) (0,2)	UniPC 8 calls
G3	(0,3) (1,2) (2,1)	UniPC 8 calls

Outer: AR over 4 groups of 4×4 block-DCT coefficients. Inner: a conditional diffusion ODE samples each group's whole 8×8 field (one axial transformer, exact prefix cache).

A1 freq-AR

9.46

FID-50K

params 11.1M
recall 0.413
46 serial calls
train 5.77 H100-h

R0 DCTdiff

8.68

FID-50K

params 44.2M
recall 0.440
15 serial calls
train 3.34 H100-h

Both: validated UniPC samplers, same q90 evaluator

NOT YET WON

- **46 vs 15** serial evaluations
- **1.84×** latency, 1.36× GPU-s (H100)
- **1.72×** training H100-h for this FID
- **Recall gap** grows G1→G3

Open question Is explicit frequency causality a distinct, useful inductive bias — and can its per-field generator be made cheap enough to win the quality–compute frontier, not just the parameter count?

FFHQ64-q90 (70k), EDM Inception FID-50K + k=3 P/R; A1 a1_d256 250k EMA, R0 our DCTdiff port 250k EMA; latency: isolated H100 NVL bench (bs 500 / bs 16). efficiency/phase2_verdict.json, bench*.json